ANDROID MOBILE APPLICATION DEVELOPMENT

40 Viva Questions and Answers

Simple, exam-friendly answers for practical viva preparation

Topics covered

Android fundamentals • Architecture • Activities and lifecycle • Layouts and UI widgets • Intents • SQLite and CRUD • Toast and AlertDialog • Gradle, APK and debugging

Unit 1: Introduction to Mobile Application Development

1. What is mobile application development, and how is it different from web application development?

Answer: Mobile application development is the process of creating software for smartphones and tablets. Mobile apps are usually installed on a device and can access features such as the camera and GPS. Web applications run in a web browser and generally do not need to be installed.

2. What is the Android operating system, and what are its major features?

Answer: Android is an open-source, Linux-based operating system developed by Google for mobile devices. Its major features include a user-friendly interface, multitasking, Wi-Fi and Bluetooth support, GPS, notifications, app support and security features.

3. Explain the architecture of the Android operating system with its different layers.

Answer: Android architecture is commonly described in layers: (1) Applications – apps such as Phone and Camera; (2) Application Framework – services such as Activity Manager and Notification Manager; (3) Android Runtime (ART) – runs app code; (4) Native Libraries – libraries such as SQLite and graphics libraries; and (5) Linux Kernel – handles drivers, memory, processes and security.

4. What is Android Studio, and why is it used for developing Android applications?

Answer: Android Studio is the official Integrated Development Environment (IDE) for Android development. It is used to write Java or Kotlin code, design layouts, build apps, run emulators, test applications and debug errors.

5. What is an Android SDK, and what is its importance in application development?

Answer: SDK stands for Software Development Kit. It contains the APIs, libraries and tools needed to develop, compile, test and debug Android applications.

6. What is an Android Virtual Device (AVD), and how is it used for testing applications?

Answer: AVD stands for Android Virtual Device. It is a configuration for a virtual Android device, including its Android version, screen size and hardware settings. Developers use it to test apps on different device configurations.

7. What is an Emulator, and how does it help developers during application development?

Answer: An emulator is software that simulates an Android device on a computer. It lets developers install, run and test apps without needing a physical phone.

Unit 2: Activity and Android Components

8. What is an Activity in Android, and how does it represent a screen in an application?

Answer: An Activity is an Android application component that provides a screen with which the user can interact. Examples include Login, Registration and Home screens.

9. Explain the Activity Lifecycle and describe its important callback methods.

Answer: The Activity Lifecycle describes the states an Activity passes through. Important callbacks are onCreate() (initial setup), onStart() (becomes visible), onResume() (becomes interactive), onPause() (loses focus), onStop() (no longer visible), onRestart() (starts again after stopping) and onDestroy() (being destroyed).

10. What is the purpose of the onCreate() method in Android?

Answer: onCreate() is called when an Activity is first created. It is commonly used to initialize components, set up the user interface with setContentView(), and prepare required data.

11. What is the difference between onStart() and onResume() methods?

Answer: onStart() makes the Activity visible to the user. onResume() runs after onStart() and brings the Activity to the foreground so the user can interact with it.

12. What is the AndroidManifest.xml file, and why is it required in an Android project?

Answer: AndroidManifest.xml is an application configuration file. It declares important information such as app components, permissions, features and application settings so Android can understand how the app should run.

13. What is the purpose of declaring an Activity inside the AndroidManifest.xml file?

Answer: Declaring an Activity registers it with the Android system. This allows Android to identify and launch the Activity when it is requested, subject to the app's configuration and exported-component rules.

Unit 3: Android Layouts and UI Components

14. What is a Layout in Android, and why is it required for designing user interfaces?

Answer: A Layout is a ViewGroup that controls how UI elements are positioned and sized on the screen. It is used to organize items such as TextViews, EditTexts and Buttons.

15. What is LinearLayout, and how does it arrange UI components on the screen?

Answer: LinearLayout arranges its child views in one direction: either in a row or in a column. Its orientation property controls the direction.

16. What is the difference between horizontal and vertical orientation in LinearLayout?

Answer: Horizontal orientation places components side by side, from left to right. Vertical orientation places components one below another, from top to bottom.

17. What is RelativeLayout, and how does it position UI components?

Answer: RelativeLayout positions views in relation to the parent layout or to other views. For example, a Button can be placed below a TextView. ConstraintLayout is generally preferred for many modern complex layouts.

18. What is ConstraintLayout, and what are its advantages in Android UI design?

Answer: ConstraintLayout positions views by defining constraints between views and the parent. It supports flexible layouts and can reduce deeply nested view hierarchies.

19. What is the difference between match_parent and wrap_content?

Answer: match_parent makes a view use the available size of its parent in that dimension. wrap_content makes the view use only the space needed to display its content, subject to constraints.

20. What is the purpose of TextView in Android applications?

Answer: TextView is a UI widget used to display text, such as headings, labels, instructions or messages.

21. What is EditText, and how is it used to accept input from users?

Answer: EditText is a UI widget that allows users to enter or edit text. It is commonly used for names, email addresses, passwords and search fields.

22. What is the purpose of a Button widget in Android?

Answer: A Button is a clickable UI element that lets the user trigger an action, such as Login, Submit, Save or Register.

23. What is an OnClickListener, and how is it used to handle button click events?

Answer: OnClickListener is an event listener used to respond when a view is clicked. A click listener is attached to a Button, and its callback contains the action to perform.

Unit 4: Intent and Activity Navigation

24. What is an Intent in Android, and why is it used?

Answer: An Intent is a messaging object used to request an action from another Android component. It is commonly used to start an Activity, start a service or request an action from another app.

25. Explain the difference between Explicit Intent and Implicit Intent with suitable examples.

Answer: An Explicit Intent names the component to open, such as opening the HomeActivity in the same app. An Implicit Intent describes an action without naming a specific component, such as asking Android to open a web page or share text.

26. How can you navigate from one Activity to another using an Intent?

Answer: Create an Intent with the current Activity and target Activity, then call startActivity(intent). The target Activity must be declared in the manifest when required.

27. What is the purpose of passing data between two Activities?

Answer: Passing data allows one Activity to share information with another, such as sending a username from a Login screen to a Home screen. Extras can be added with putExtra() and read from the receiving Intent.

Unit 5: SQLite Database and CRUD Operations

28. What is SQLite, and why is it used in Android applications?

Answer: SQLite is a lightweight relational database that can store structured data locally on an Android device. It works offline and does not require a separate database server.

29. What is the difference between SQLite and SharedPreferences?

Answer: SQLite stores structured data in tables and supports queries. SharedPreferences stores small key-value settings, such as a theme choice or a simple login flag. Modern Android projects may also use DataStore for preferences.

30. What is a database in Android, and how can it be created using SQLite?

Answer: A database is an organized collection of data. In traditional Android SQLite development, a database can be created by extending SQLiteOpenHelper and defining the table-creation SQL in onCreate().

31. What is the purpose of the SQLiteOpenHelper class?

Answer: SQLiteOpenHelper helps create, open and manage an SQLite database. Developers commonly override onCreate() to create tables and onUpgrade() to handle database schema changes.

32. What are the important methods used to perform CRUD operations in SQLite?

Answer: CRUD means Create, Read, Update and Delete. Common SQLiteDatabase methods include insert() for adding records, query() or rawQuery() for reading, update() for modifying and delete() for removing records.

33. What is the difference between INSERT, UPDATE, DELETE, and SELECT operations?

Answer: INSERT adds a new record. UPDATE changes an existing record. DELETE removes records. SELECT retrieves records from a database.

Unit 6: Android Messages and UI Utilities

34. What is a Toast message, and how can it be displayed in an Android application?

Answer: A Toast is a brief message displayed to the user without blocking interaction. In Java, it can be shown with Toast.makeText(context, "Message", Toast.LENGTH_SHORT).show().

35. What is the purpose of AlertDialog in Android applications?

Answer: AlertDialog displays a message or asks the user to make a decision. For example, it can ask the user to confirm before deleting a record or exiting an app.

36. What is the difference between dp, sp, and px in Android?

Answer: dp means density-independent pixels and is used for layout dimensions. sp means scale-independent pixels and is recommended for text sizes because it respects font scaling. px means physical screen pixels.

Unit 7: Build System and Testing

37. What is the purpose of the Gradle build system in Android Studio?

Answer: Gradle automates the app build process. It manages dependencies, compiles source code, processes resources, runs build tasks and packages the app.

38. What is APK, and why is it generated during Android application development?

Answer: APK stands for Android Package Kit. It is an installable package format containing the compiled app code, resources, manifest and other required files.

39. What is application debugging, and how can Logcat help developers identify errors?

Answer: Debugging is the process of finding and fixing problems in an application. Logcat displays logs, warnings, exceptions and stack traces that help developers understand what happened while the app was running.

40. Explain the complete process of developing, testing, and running an Android application using Android Studio.

Answer: Install Android Studio and the required SDK; create a project; choose Java or Kotlin; design the UI; write the application logic; configure the manifest and resources; create or connect a test device; run the app; test its features; inspect errors in Logcat; fix issues; and build an APK or app bundle when ready.

Quick Revision

- **Android:** Open-source, Linux-based mobile operating system.
- **AVD:** Android Virtual Device configuration used by the emulator.
- **Activity:** A screen that provides a user interface.
- **CRUD:** Create, Read, Update, Delete.
- **SQLite:** Local relational database.
- **APK:** Android installable application package.
- **dp:** Layout dimensions; **sp:** Text size; **px:** Physical pixels.
- **Logcat:** Displays app logs and errors.

Viva tip: Understand each definition, its purpose and one example. Answer calmly in two to four sentences. If you do not know an answer, explain the part you understand instead of guessing.